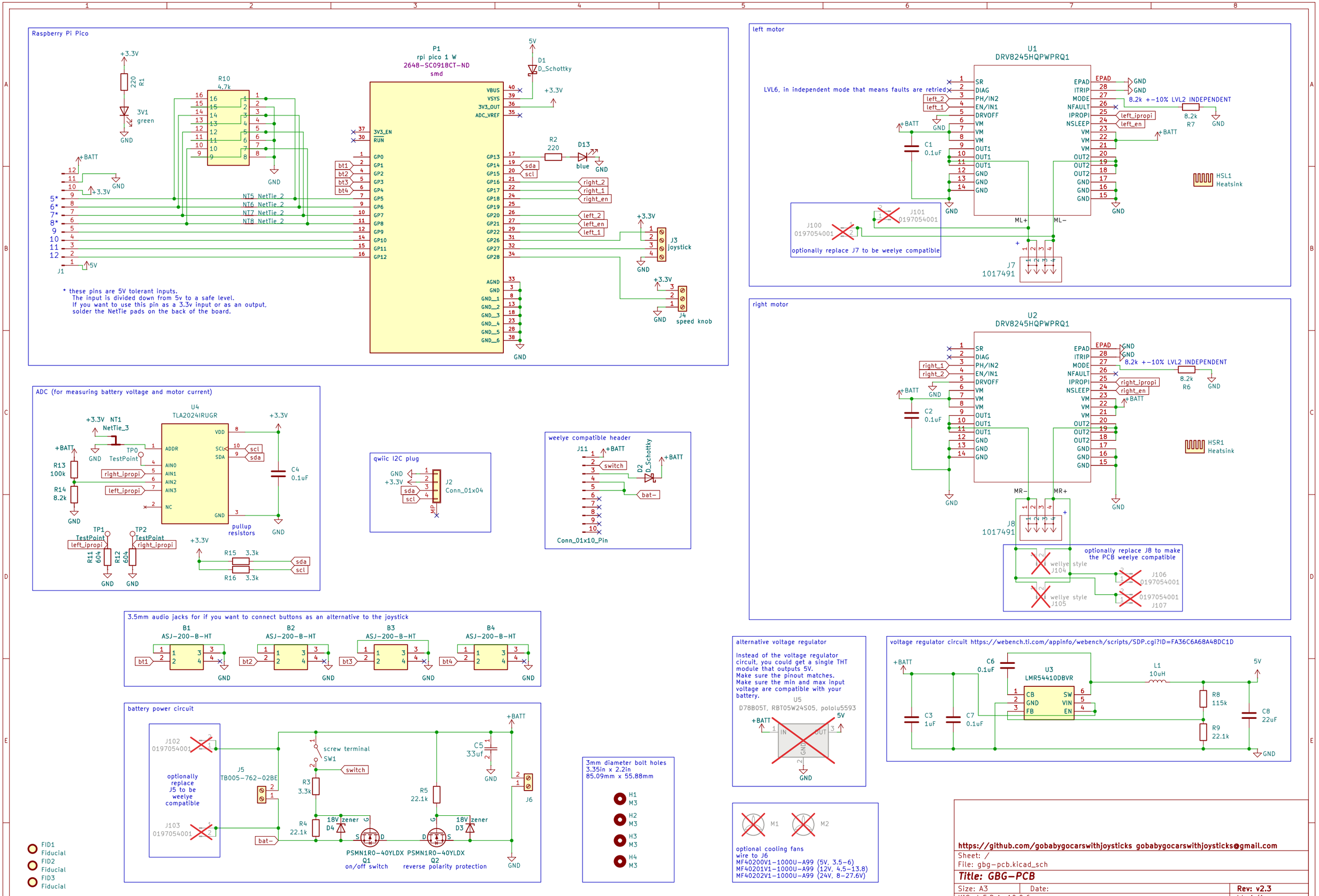
Raspberry Pi Pico
P1
rpi pico 1 W
2648-SC0918CT-ND
smd
* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.
left motor
U1
DRV8245HQPWPRQ1
LVL6, in independent mode that means faults are retried
8.2k +-10% LVL2 INDEPENDENT
HSL1
Heatsink
J100
0197054001
J101
0197054001
optionally replace J7 to be weelye compatible
J7
1017491
right motor
U2
DRV8245HQPWPRQ1
8.2k +-10% LVL2 INDEPENDENT
HSR1
Heatsink
J8
1017491
optionally replace J8 to make the PCB weelye compatible
wellye style
J104
wellye style
J105
J106
0197054001
0197054001
J107
ADC (for measuring battery voltage and motor current)
U4
TLA2024IRUGR
pullup resistors
qwiic I2C plug
J2
Conn_01x04
weelye compatible header
J11
Conn_01x10_Pin
3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1
ASJ-200-B-HT
B2
ASJ-200-B-HT
B3
ASJ-200-B-HT
B4
ASJ-200-B-HT
battery power circuit
J102
0197054001
optionally replace J5 to be weelye compatible
J103
0197054001
J5
TB005-762-02BE
screw terminal
SW1
PSMN1R0-40YLDX
Q1
on/off switch
PSMN1R0-40YLDX
Q2
reverse polarity protection
J6
3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1
M3
H2
M3
H3
M3
H4
M3
alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593
voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=FA36C6A68A48DC1D
U3
LMR54410DBVR
M1
M2
optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6)
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title: GBG-PCB
Size: A3
Date:
Rev: v2.3
KiCad E.D.A. 10.0.6
Id: 1/1